

Adham Aly

Oakville, ON | adham.alysaleh@gmail.com | linkedin.com/in/adhamalysaleh | github.com/adham-aly | adhamaly.com

SUMMARY

AI/ML Engineer and Computer Science student at the University of Toronto. Builds systems on large language models: multi-agent coding workflows, cost-based model routing, and context compaction for long-running agents.

EDUCATION

University of Toronto	Toronto, ON
Honours Bachelor of Computer Science (BCS), Computer Science Specialist (AI Focus)	Sept. 2026 – 2030 (Expected)
White Oaks Secondary School — International Baccalaureate (IB) Diploma	Sept. 2022 – July 2026

TECHNICAL SKILLS

Languages: C, C++, TypeScript, JavaScript, Python, SQL, x86 Assembly, HTML/CSS

Frameworks & Libraries: Next.js, React, Node.js

AI & Data: LLM agents (Claude Code, Codex), multi-agent orchestration, Jev, Ollama, Hugging Face, Prophet

Developer Tools: Docker, Git, Linux, QEMU, GCC, Make/CMake

Concepts: Data Structures & Algorithms, Operating Systems, Machine Learning, REST APIs, Containerization

PROJECTS

Multi-Agent Coding Workflow — Open-Source Agent Skill | Claude Code, Codex, Cursor, Jev

- Built an open-source agent skill for Claude Code, Codex, and Cursor that runs a coding task through planning, implementation, and review, each phase owned by an orchestrator agent with up to 4 subagents
- Routes each subagent to a model and reasoning effort with Jev, TypeSafe's fast decision model, sending lookups and small edits to a cheap model and design work to a frontier model at high effort, cutting the cost of a run
- Uses the same decision model to choose when to compact an agent's context, keeping decisions, blockers, and next steps and dropping stale tool output, so long runs stay inside the context window

WMOJ — Competitive Programming Online Judge | TypeScript, Next.js, PostgreSQL, Docker

- Co-founded WMOJ, an open-source competitive-programming judge with 80 problems, timed contests, and live leaderboards, used by 3+ Halton District School Board schools for the WOSS Dual Olympiad and TriOlympiad
- Built the grading server that runs untrusted Python and C++ submissions inside an nsjail sandbox with a seccomp filter that blocks network access and process tampering, plus CPU, memory, and output limits
- Found and fixed a bug that had graded 94 real timeouts as wrong answers, writing a small C program that records each run's exact CPU time and memory use
- Ran a security review of the web app that fixed 70 defects, adding 64 PostgreSQL row-level security policies and closing holes that exposed answer keys and student code and let anyone forge submissions

Scire — Peer-to-Peer Tutoring LMS | Next.js, React, TypeScript

- Built Scire, a full-stack tutoring learning-management system in Next.js, React, and TypeScript, with tutor-tutee matching and scheduling, session tracking, and admin dashboards
- In active use at a high-school tutoring club; seeking school-board approval to expand it district-wide

EXPERIENCE

President, Computer Science Club & Director, WOSS TriOlympiad	Sept. 2024 – July 2026
White Oaks Secondary School	Oakville, ON

- Directed a 15+ person executive team running computer-science and competitive-programming events, including the WOSS Dual Olympiad, owning logistics, marketing, and sponsorship outreach
- Taught weekly data-structures, algorithms, and software-engineering lessons to a class of 50+ students, building their competitive-programming and problem-solving skills
- Directed the WOSS TriOlympiad, a free three-day physics, math, and computer-science competition drawing 100+ students across the school board, and co-built its registration site (triolympiad.ca) in Next.js